CHAPTER 31 SUMMARY

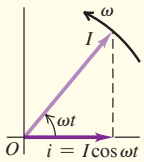
Phasors and alternating current: An alternator or ac source produces an emf that varies sinusoidally with time. A sinusoidal voltage or current can be represented by a phasor, a vector that rotates counterclockwise with constant angular velocity ω equal to the angular frequency of the sinusoidal quantity. Its projection on the horizontal axis at any instant represents the instantaneous value of the quantity.

For a sinusoidal current, the rectified average and rms (root-mean-square) currents are proportional to the current amplitude I. Similarly, the rms value of a sinusoidal voltage is proportional to the voltage amplitude V. (See Example 31.1.)

$$I_{\text{rav}} = \frac{2}{\pi} I = 0.637I \quad (31.3)$$

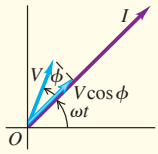
$$I_{\text{rms}} = \frac{I}{\sqrt{2}} \quad (31.4)$$

$$V_{\text{rms}} = \frac{V}{\sqrt{2}} \quad (31.5)$$



Voltage, current, and phase angle: In general, the instantaneous voltage $v = V\cos(\omega t + \phi)$ between two points in an ac circuit is not in phase with the instantaneous current passing through those points. The quantity ϕ is called the phase angle of the voltage relative to the current.

$$i = I\cos\omega t \quad (31.2)$$

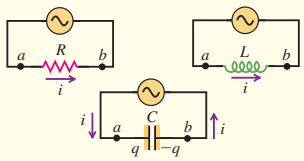


Resistance and reactance: The voltage across a resistor R is in phase with the current. The voltage across an inductor L leads the current by 90° ($\phi = +90°$), while the voltage across a capacitor C lags the current by 90° ($\phi = -90°$). The voltage amplitude across each type of device is proportional to the current amplitude I. An inductor has inductive reactance $X_L = \omega L$, and a capacitor has capacitive reactance $X_C = 1/\omega C$. (See Examples 31.2 and 31.3.)

$$V_R = IR \quad (31.7)$$

$$V_L = IX_L \quad (31.13)$$

$$V_C = IX_C \quad (31.19)$$

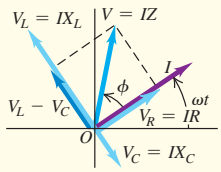


Impedance and the *L-R-C* series circuit: In a general ac circuit, the voltage and current amplitudes are related by the circuit impedance Z. In an L-R-C series circuit, the values of L, R, C, and the angular frequency ω determine the impedance and the phase angle ϕ of the voltage relative to the current. (See Examples 31.4 and 31.5.)

$$V = IZ \quad (31.22)$$

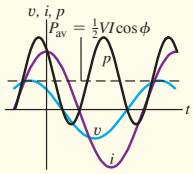
$$Z = \sqrt{R^2 + [\omega L - (1/\omega C)]^2} \quad (31.23)$$

$$\tan\phi = \frac{\omega L - 1/\omega C}{R} \quad (31.24)$$



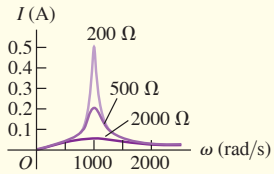
Power in ac circuits: The average power input P_{av} to an ac circuit depends on the voltage and current amplitudes (or, equivalently, their rms values) and the phase angle ϕ of the voltage relative to the current. The quantity $\cos\phi$ is called the power factor. (See Examples 31.6 and 31.7.)

$$P_{\text{av}} = \tfrac{1}{2}VI\cos\phi = V_{\text{rms}}I_{\text{rms}}\cos\phi \quad (31.31)$$



Resonance in ac circuits: In an L-R-C series circuit, the current becomes maximum and the impedance becomes minimum at an angular frequency called the resonance angular frequency. This phenomenon is called resonance. At resonance the voltage and current are in phase, and the impedance Z is equal to the resistance R. (See Example 31.8.)

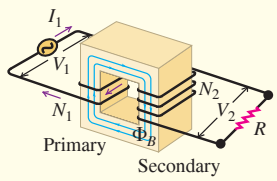
$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (31.32)$$



Transformers: A transformer is used to transform the voltage and current levels in an ac circuit. In an ideal transformer with no energy losses, if the primary winding has N_1 turns and the secondary winding has N_2 turns, the amplitudes (or rms values) of the two voltages are related by Eq. (31.35). The amplitudes (or rms values) of the primary and secondary voltages and currents are related by Eq. (31.36). (See Example 31.9.)

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \quad (31.35)$$

$$V_1 I_1 = V_2 I_2 \quad (31.36)$$



Chapter 31 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 31.2 and 31.3 (Section 31.2) before attempting these problems.

VP31.3.1 An inductor with $L = 2.50$ mH is part of the circuit for an aviation radio beacon that operates at 108 MHz. Find (a) the inductive reactance of the inductor and (b) the amplitude of the current in the inductor if the amplitude of the voltage across the inductor is 4.20 kV.

VP31.3.2 An ac circuit includes a resistor with $R = 125\ \Omega$ connected in series with a capacitor with $C = 7.00\ \mu$F. The current through the resistor and capacitor is $i = (2.40 \times 10^{-3}\ \text{A})\cos[(1.75 \times 10^3\ \text{rad/s})t]$. Find (a) the voltage across the resistor as a function of time, (b) the capacitive reactance of the capacitor, and (c) the voltage across the capacitor as a function of time.

VP31.3.3 An ac circuit includes a 155 Ω resistor in series with a 8.00 μF capacitor. The current in the circuit has amplitude 4.00×10^{-3} A. (a) Find the frequency for which the capacitive reactance equals the resistance. (b) At this frequency, what are the amplitudes of the voltages across the resistor and capacitor? (c) At this frequency, when the voltage across the resistor is maximum, what is the voltage across the capacitor? When the voltage across the capacitor is maximum, what is the voltage across the resistor?

VP31.3.4 A 116 Ω resistor and an unknown capacitor are connected in series to an ac source with angular frequency 5.10×10^3 rad/s. The amplitude of the resistor voltage is 2.45 V, and the current in the circuit has its maximum positive value at $t = 0$. Find (a) the current amplitude, (b) the current at $t = 3.50$ ms, (c) the maximum voltage across the capacitor, and (d) the capacitance.

Be sure to review EXAMPLES 31.4 and 31.5 (Section 31.3) before attempting these problems.

VP31.5.1 In the series circuit of Fig. 31.12a the resistance is 255 Ω, the inductance is 45.0 mH, the capacitance is 0.400 μF, and the source has voltage amplitude 55.0 V. The inductive reactance is equal to the resistance. Find (a) the frequency of the source, (b) the impedance, (c) the current amplitude, (d) the phase angle, and (e) whether the voltage leads or lags the current.

VP31.5.2 In the series circuit of Fig. 31.12a the source has voltage amplitude 20.0 V and angular frequency 5.40×10^3 rad/s, the inductance is 6.50 mH, the capacitance is 0.600 μF, and the impedance is 474 Ω. Find (a) the resistance, (b) the current amplitude, (c) the resistor voltage amplitude, (d) the inductor voltage amplitude, and (e) the capacitor voltage amplitude.

VP31.5.3 The source in an *L-R-C* series circuit has angular frequency 1.20×10^4 rad/s. The capacitance is 0.965 μF and the resistance is 65.0 Ω. If you want the source voltage to lead the current by 15.0°, find (a) the required value of $X_L - X_C$ and (b) the required inductance.

VP31.5.4 The current in an *L-R-C* series circuit has amplitude 0.120 A and angular frequency 8.00×10^3 rad/s, and it has its maximum positive value at $t = 0$. The resistance is 95.0 Ω, the inductance is 6.50 mH, and the capacitance is 0.440 μF. For the resistor, inductor, and capacitor, find (a) the voltage amplitudes and (b) the instantaneous voltages at $t = 0.305$ ms.

Be sure to review EXAMPLES 31.6 and 31.7 (Section 31.4) before attempting these problems.

VP31.7.1 In operation an electric toaster has an rms current of 4.13 A when plugged into a wall socket with rms voltage 230 V. Find the toaster's (a) average power, (b) maximum instantaneous power, and (c) resistance.

VP31.7.2 An *L-R-C* series circuit has a source with voltage amplitude 35.0 V and angular frequency 1.30×10^3 rad/s. The resistance is 275 Ω, the inductance is 82.3 mH, and the capacitance is 1.10 μF. Find (a) the inductive and capacitive reactances, (b) the phase angle, and (c) the power factor.

VP31.7.3 For the circuit in the preceding problem, find (a) the impedance, (b) the current amplitude, and (c) the average power delivered to the resistor.

VP31.7.4 The power factor of an *L-R-C* series circuit is 0.800. The angular frequency is 1.30×10^4 rad/s, the inductance is 7.50 mH, and the capacitance is 0.440 μF. Find (a) the inductive and capacitive reactances, (b) the phase angle, and (c) the resistance.

BRIDGING PROBLEM An Alternating-Current Circuit

A series circuit like the circuit in Fig. 31.12a consists of a 1.50 mH inductor, a 125 Ω resistor, and a 25.0 nF capacitor connected across an ac source having an rms voltage of 35.0 V and variable frequency. (a) At what angular frequencies will the current amplitude be equal to $\frac{1}{3}$ of its maximum possible value? (b) At the frequencies in part (a), what are the current amplitude and the voltage amplitude across each circuit element (including the ac source)?

SOLUTION GUIDE

IDENTIFY and SET UP

1. The maximum current amplitude occurs at the resonance angular frequency. This problem concerns the angular frequencies at which the current amplitude is one-third of that maximum.
2. Choose the equation that will allow you to find the angular frequencies in question, and choose the equations that you'll then use to find the current and voltage amplitudes at each angular frequency.

EXECUTE

3. Find the impedance at the angular frequencies in part (a); then solve for the values of angular frequency.
4. Find the voltage amplitude across the source and the current amplitude for each of the angular frequencies in part (a). (*Hint:* Be careful to distinguish between *amplitude* and *rms value*.)
5. Use the results of steps 3 and 4 to find the reactances at each angular frequency. Then calculate the voltage amplitudes for the resistor, inductor, and capacitor.

EVALUATE

6. Are any voltage amplitudes greater than the voltage amplitude of the source? If so, does this mean your results are in error?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q31.1 Household electric power in most of western Europe is supplied at 230 V, rather than the 120 V that is standard in the United States and Canada. What are the advantages and disadvantages of each system?

Q31.2 The current in an ac power line changes direction 120 times per second, and its average value is zero. Explain how it is possible for power to be transmitted in such a system.

Q31.3 In an ac circuit, why is the average power for an inductor and a capacitor zero, but not for a resistor?

Q31.4 Equation (31.14) was derived by using the relationship $i = dq/dt$ between the current and the charge on the capacitor. In Fig. 31.8a the positive counterclockwise current increases the charge on the capacitor. When the charge on the left plate is positive but decreasing in time, is $i = dq/dt$ still correct or should it be $i = -dq/dt$? Is $i = dq/dt$ still correct when the right-hand plate has positive charge that is increasing or decreasing in magnitude? Explain.

Q31.5 Fluorescent lights often use an inductor, called a ballast, to limit the current through the tubes. Why is it better to use an inductor rather than a resistor for this purpose?

Q31.6 Equation (31.9) says that $v_{ab} = L\,di/dt$ (see Fig. 31.7a). Using Faraday's law, explain why point a is at higher potential than point b when i is in the direction shown in Fig. 31.7a and is increasing in magnitude. When i is counterclockwise and decreasing in magnitude, is $v_{ab} = L\,di/dt$ still correct, or should it be $v_{ab} = -L\,di/dt$? Is $v_{ab} = L\,di/dt$ still correct when i is clockwise and increasing or decreasing in magnitude? Explain.

Q31.7 Is it possible for the power factor of an *L-R-C* series ac circuit to be zero? Justify your answer on *physical* grounds.

Q31.8 In an *L-R-C* series circuit, can the instantaneous voltage across the capacitor exceed the source voltage at that same instant? Can this be true for the instantaneous voltage across the inductor? Across the resistor? Explain.

Q31.9 In an *L-R-C* series circuit, what are the phase angle ϕ and power factor $\cos\phi$ when the resistance is much smaller than the inductive or capacitive reactance and the circuit is operated far from resonance? Explain.

Q31.10 When an *L-R-C* series circuit is connected across a 230 V ac supply, the voltage rating of the capacitor may be exceeded even if it is rated at 400 V. How can this be?

Q31.11 In Example 31.6 (Section 31.4), a hairdryer is treated as a pure resistor. But because there are coils in the heating element and in the motor that drives the blower fan, a hairdryer also has inductance. Qualitatively, does including an inductance increase or decrease the values of R, I_{rms}, and P?

Q31.12 A light bulb and a parallel-plate capacitor with air between the plates are connected in series to an ac source. What happens to the brightness of the bulb when a dielectric is inserted between the plates of the capacitor? Explain.

Q31.13 A coil of wire wrapped on a hollow tube and a light bulb are connected in series to an ac source. What happens to the brightness of the bulb when an iron rod is inserted in the tube?

Q31.14 A circuit consists of a light bulb, a capacitor, and an inductor connected in series to an ac source. What happens to the brightness of the bulb when the inductor is omitted? When the inductor is left in the circuit but the capacitor is omitted? Explain.

Q31.15 A circuit consists of a light bulb, a capacitor, and an inductor connected in series to an ac source. Is it possible for both the capacitor and the inductor to be removed and the brightness of the bulb to remain the same? Explain.

Q31.16 Can a transformer be used with dc? Explain. What happens if a transformer designed for 230 V ac is connected to a 230 V dc supply?

Q31.17 An ideal transformer has N_1 windings in the primary and N_2 windings in its secondary. If you double only the number of secondary windings, by what factor does (a) the voltage amplitude in the secondary change, and (b) the effective resistance of the secondary circuit change?

Q31.18 An inductor, a capacitor, and a resistor are all connected in series across an ac source. If the resistance, inductance, and capacitance are all doubled, by what factor does each of the following quantities change? Indicate whether they increase or decrease: (a) the resonance angular frequency; (b) the inductive reactance; (c) the capacitive reactance. (d) Does the impedance double?

Q31.19 You want to double the resonance angular frequency of an *L-R-C* series circuit by changing only the *pertinent* circuit elements all by the same factor. (a) Which ones should you change? (b) By what factor should you change them?

EXERCISES

Section 31.1 Phasors and Alternating Currents

31.1 • You have a special light bulb with a *very* delicate wire filament. The wire will break if the current in it ever exceeds 1.40 A, even for an instant. What is the largest root-mean-square current you can run through this bulb?

31.2 • The voltage across the terminals of an ac power supply varies with time according to Eq. (31.1). The voltage amplitude is $V = 43.0$ V. What are (a) the root-mean-square potential difference V_{rms} and (b) the average potential difference V_{av} between the two terminals of the power supply?

Section 31.2 Resistance and Reactance

31.3 • An inductor with $L = 9.50$ mH is connected across an ac source that has voltage amplitude 45.0 V. (a) What is the phase angle ϕ for the source voltage relative to the current? Does the source voltage lag or lead the current? (b) What value for the frequency of the source results in a current amplitude of 3.90 A?

31.4 • A capacitor is connected across an ac source that has voltage amplitude 58.5 V and frequency 83.5 Hz. (a) What is the phase angle ϕ for the source voltage relative to the current? Does the source voltage lag or lead the current? (b) What is the capacitance C of the capacitor if the current amplitude is 5.50 A?

31.5 • (a) What is the reactance of a 3.00 H inductor at a frequency of 80.0 Hz? (b) What is the inductance of an inductor whose reactance is 120 Ω at 80.0 Hz? (c) What is the reactance of a 4.00 μF capacitor at a frequency of 80.0 Hz? (d) What is the capacitance of a capacitor whose reactance is 120 Ω at 80.0 Hz?

31.6 • A capacitance C and an inductance L are operated at the same angular frequency. (a) At what angular frequency will they have the same reactance? (b) If $L = 4.70$ mH and $C = 3.00$ μF, what is the numerical value of the angular frequency in part (a), and what is the reactance of each element?

31.7 • (a) Compute the reactance of a 0.450 H inductor at frequencies of 60.0 Hz and 600 Hz. (b) Compute the reactance of a 2.50 μF capacitor at the same frequencies. (c) At what frequency is the reactance of a 0.450 H inductor equal to that of a 2.50 μF capacitor?

31.8 • **A Radio Inductor.** You want the current amplitude through a 4.50 mH inductor (part of the circuitry for a radio receiver) to be 2.10 mA when a sinusoidal voltage with amplitude 12.0 V is applied across the inductor. What frequency is required?

31.9 •• A 0.200 H inductor is connected in series with a 81.5 Ω resistor and an ac source. The voltage across the inductor is $v_L = -(13.0\text{ V})\sin[(474\text{ rad/s})t]$. (a) Derive an expression for the voltage v_R across the resistor. (b) What is v_R at $t = 1.93$ ms?

31.10 •• A 250 Ω resistor is connected in series with a 4.80 μF capacitor and an ac source. The voltage across the capacitor is $v_C = (7.60\text{ V})\sin[(120\text{ rad/s})t]$. (a) Determine the capacitive reactance of the capacitor. (b) Derive an expression for the voltage v_R across the resistor.

31.11 •• A 150 Ω resistor is connected in series with a 0.250 H inductor and an ac source. The voltage across the resistor is $v_R = (3.80\text{ V})\cos[(720\text{ rad/s})t]$. (a) Derive an expression for the circuit current. (b) Determine the inductive reactance of the inductor. (c) Derive an expression for the voltage v_L across the inductor.

Section 31.3 The *L-R-C* Series Circuit

31.12 • You have a 200 Ω resistor, a 0.400 H inductor, and a 6.00 μF capacitor. Suppose you take the resistor and inductor and make a series circuit with a voltage source that has voltage amplitude 30.0 V and an angular frequency of 250 rad/s. (a) What is the impedance of the circuit? (b) What is the current amplitude? (c) What are the voltage amplitudes across the resistor and across the inductor? (d) What is the phase angle ϕ of the source voltage with respect to the current? Does the source voltage lag or lead the current? (e) Construct the phasor diagram.

31.13 • The resistor, inductor, capacitor, and voltage source described in Exercise 31.12 are connected to form an *L-R-C* series circuit. (a) What is the impedance of the circuit? (b) What is the current amplitude? (c) What is the phase angle of the source voltage with respect to the current? Does the source voltage lag or lead the current? (d) What are the voltage amplitudes across the resistor, inductor, and capacitor? (e) Explain how it is possible for the voltage amplitude across the capacitor to be greater than the voltage amplitude across the source.

31.14 •• A 220 Ω resistor, 0.875 H inductor, and 7.00 μF capacitor are connected in series across a voltage source that has voltage amplitude 30.5 V and an angular frequency of 260 rad/s. (a) What are v, v_R, v_L, and v_C at $t = 20.5$ ms? Compare $v_R + v_L + v_C$ to v at this instant. (b) What are V_R, V_L, and V_C? Compare V to $V_R + V_L + V_C$. Explain why these two quantities are not equal.

31.15 • In an *L-R-C* series circuit, the rms voltage across the resistor is 30.0 V, across the capacitor it is 90.0 V, and across the inductor it is 50.0 V. What is the rms voltage of the source?

31.16 •• An *L-R-C* series circuit has voltage amplitudes $V_L = 180$ V, $V_C = 120$ V, and $V_R = 160$ V. At time t the instantaneous voltage across the inductor is 80.0 V. At this instant, what is the voltage across the capacitor and across the resistor?

31.17 • An *L-R-C* series circuit has source voltage amplitude $V = 240$ V, and the voltage amplitudes for the inductor and capacitor are $V_L = 310$ V and $V_C = 180$ V. What is the phase angle ϕ?

Section 31.4 Power in Alternating-Current Circuits

31.18 •• A resistor with $R = 295$ Ω and an inductor are connected in series across an ac source that has voltage amplitude 495 V. The rate at which electrical energy is dissipated in the resistor is 210 W. What is (a) the impedance Z of the circuit; (b) the amplitude of the voltage across the inductor; (c) the power factor?

31.19 • The power of a certain DVD player operating at 230 V rms is 16.0 W. Assuming that the DVD player behaves like a pure resistor, find (a) the maximum instantaneous power; (b) the rms current; (c) the resistance of this player.

31.20 •• In an *L-R-C* series circuit, the components have the following values: $L = 20.0$ mH, $C = 140$ nF, and $R = 350$ Ω. The generator has an rms voltage of 230 V and a frequency of 1.25 kHz. Determine (a) the power supplied by the generator and (b) the power dissipated in the resistor.

31.21 •• (a) Use Figs. 31.12b and 31.13c to show that for a series *L-R-C* circuit, $V\cos\phi = IR$. Use this result in Eq. (31.31) to show that the average power delivered by the source is $P_{av} = \frac{1}{2}I^2R$. Therefore the average power delivered by the source is equal to the average power consumed by the resistor. This means the average power for the capacitor and for the inductor is zero. These circuit elements consume electrical energy during the part of the current oscillations when they are storing energy, but all the stored energy in each is then released during another part of the current oscillations. (b) In an *L-R-C* series circuit the amplitude of the source voltage is 230 V, the source voltage leads the current by 53.1°, and the average power supplied by the source is 110.0 W. What is the resistance R of the resistor in the circuit?

31.22 •• A circuit has an ac voltage source and a resistor and capacitor connected in series. There is no inductor. The ac voltage source has voltage amplitude 900 V and angular frequency $\omega = 20.0$ rad/s. The voltage amplitude across the capacitor is 500 V. The resistor has resistance $R = 300$ Ω. (a) What is the voltage amplitude across the resistor? (b) What is the capacitance C of the capacitor? (c) Does the source voltage lag or lead the current? (d) What is the average rate at which the ac source supplies electrical energy to the circuit?

31.23 • An *L-R-C* series circuit with $L = 0.121$ H, $R = 243$ Ω, and $C = 7.26$ μF carries an rms current of 0.454 A with a frequency of 401 Hz. (a) What are the phase angle and power factor for this circuit? (b) What is the impedance of the circuit? (c) What is the rms voltage of the source? (d) What average power is delivered by the source? (e) What is the average rate at which electrical energy is converted to thermal energy in the resistor? (f) What is the average rate at which electrical energy is dissipated (converted to other forms) in the capacitor? (g) In the inductor?

31.24 •• An *L-R*-C series circuit is connected to a 120 Hz ac source that has $V_{rms} = 73.0$ V. The circuit has a resistance of 76.0 Ω and an impedance at this frequency of 119 Ω. What average power is delivered to the circuit by the source?

31.25 •• A series ac circuit contains a 290 Ω resistor, a 14 mH inductor, a 2.5 μF capacitor, and an ac power source of voltage amplitude 45 V operating at an angular frequency of 360 rad/s. (a) What is the power factor of this circuit? (b) Find the average power delivered to the entire circuit. (c) What is the average power delivered to the resistor, to the capacitor, and to the inductor?

Section 31.5 Resonance in Alternating-Current Circuits

31.26 •• In an *L-R-C* series circuit the source is operated at its resonant angular frequency. At this frequency, the reactance X_C of the capacitor is 220 Ω and the voltage amplitude across the capacitor is 590 V. The circuit has $R = 288$ Ω. What is the voltage amplitude of the source?

31.27 • **Analyzing an *L-R-C* Circuit.** You have a 210 Ω resistor, a 0.407 H inductor, a 5.10 μF capacitor, and a variable-frequency ac source with an amplitude of 2.90 V. You connect all four elements together to form a series circuit. (a) At what frequency will the current in the circuit be greatest? What will be the current amplitude at this frequency? (b) What will be the current amplitude at an angular frequency of 409 rad/s? At this frequency, will the source voltage lead or lag the current?

31.28 • An *L-R-C* series circuit is constructed using a 175 Ω resistor, a 12.5 μF capacitor, and an 8.00 mH inductor, all connected across an ac source having a variable frequency and a voltage amplitude of 25.0 V. (a) At what angular frequency will the impedance be smallest, and what is the impedance at this frequency? (b) At the angular frequency in part (a), what is the maximum current through the inductor? (c) At the angular frequency in part (a), find the potential difference across the ac source, the resistor, the capacitor, and the inductor at the instant that the current is equal to one-half its greatest positive value. (d) In part (c), how are the potential differences across the resistor, inductor, and capacitor related to the potential difference across the ac source?

31.29 • In an *L-R-C* series circuit, $R = 290\ \Omega$, $L = 0.395$ H, and $C = 6.02 \times 10^{-8}$ F. When the ac source operates at the resonance frequency of the circuit, the current amplitude is 0.505 A. (a) What is the voltage amplitude of the source? (b) What is the amplitude of the voltage across the resistor, across the inductor, and across the capacitor? (c) What is the average power supplied by the source?

31.30 • An *L-R-C* series circuit consists of a source with voltage amplitude 230 V and angular frequency 50.0 rad/s, a resistor with $R = 575\ \Omega$, an inductor with $L = 3.00$ H, and a capacitor with capacitance C. (a) For what value of C will the current amplitude in the circuit be a maximum? (b) When C has the value calculated in part (a), what is the amplitude of the voltage across the inductor?

31.31 • In an *L-R-C* series circuit, $R = 150\ \Omega$, $L = 0.710$ H, and $C = 0.0200\ \mu$F. The source has voltage amplitude $V = 150$ V and a frequency equal to the resonance frequency of the circuit. (a) What is the power factor? (b) What is the average power delivered by the source? (c) The capacitor is replaced by one with $C = 0.0370\ \mu$F and the source frequency is adjusted to the new resonance value. Then what is the average power delivered by the source?

31.32 • An *L-R-C* series circuit has $R = 400\ \Omega$, $L = 0.600$ H, and $C = 5.00 \times 10^{-8}$ F. The voltage amplitude of the source is 80.0 V. When the ac source operates at the resonance frequency of the circuit, what is the average power delivered by the source?

31.33 •• In an *L-R-C* series circuit, $L = 0.280$ H and $C = 4.00\ \mu$F. The voltage amplitude of the source is 230 V. (a) What is the resonance angular frequency of the circuit? (b) When the source operates at the resonance angular frequency, the current amplitude in the circuit is 1.70 A. What is the resistance R of the resistor? (c) At the resonance angular frequency, what are the peak voltages across the inductor, the capacitor, and the resistor?

Section 31.6 Transformers

31.34 • **Off to Europe!** A Japanese tourist takes her hairdryer to Europe, where the electrical outlets put out 230 V instead of the 100 V seen in Japan. The hairdryer puts out 2000 W at 100 V. (a) What could she do to operate her hairdryer via the 230 V supply in Europe? (b) What current will her hairdryer draw from a European outlet? (c) What resistance will her hairdryer appear to have when operated at 230 V?

31.35 • **A Step-Down Transformer.** A transformer connected to a 230 V (rms) ac line is to supply 11.5 V (rms) to a portable electronic device. The load resistance in the secondary is 5.00 Ω. (a) What should the ratio of primary to secondary turns of the transformer be? (b) What rms current must the secondary supply? (c) What average power is delivered to the load? (d) What resistance connected directly across the 230 V supply would draw the same power as the transformer? Show that this is equal to 5.00 Ω times the square of the ratio of primary to secondary turns.

31.36 • **A Step-Up Transformer.** A transformer connected to a 230 V (rms) ac line is to supply 11,000 V (rms) for a neon sign. To reduce shock hazard, a fuse is to be inserted in the primary circuit; the fuse is to blow when the rms current in the secondary circuit exceeds 30.0 mA. (a) What is the ratio of secondary to primary turns of the transformer? (b) What power must be supplied to the transformer when the rms secondary current is 30.0 mA? (c) What current rating should the fuse in the primary circuit have?

PROBLEMS

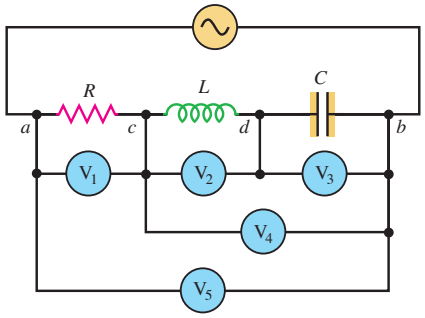
31.37 • A coil has a resistance of 44.0 Ω. At a frequency of 75.0 Hz the voltage across the coil leads the current in it by 52.8°. Determine the inductance of the coil.

31.38 •• In an *L-R-C* series circuit the phase angle is 53.0° and the source voltage lags the current. The resistance of the resistor is 300 Ω and the reactance of the capacitor is 500 Ω. The average power delivered by the source is 80.0 W. (a) What is the reactance of the inductor? (b) What is the current amplitude in the circuit? (c) What is the voltage amplitude of the source?

31.39 •• An *L-R-C* series circuit has $C = 5.00\ \mu$F, $L = 0.510$ H, and source voltage amplitude $V = 57.5$ V. The source is operated at the resonance frequency of the circuit. If the voltage across the capacitor has amplitude 77.0 V, what is the value of R for the resistor in the circuit?

31.40 •• Five infinite-impedance voltmeters, calibrated to read rms values, are connected as shown in **Fig. P31.40**. Let $R = 200\ \Omega$, $L = 0.400$ H, $C = 6.00\ \mu$F, and $V = 30.0$ V. What is the reading of each voltmeter if (a) $\omega = 200$ rad/s and (b) $\omega = 1000$ rad/s?

Figure **P31.40**



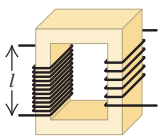
31.41 •• **CP** A parallel-plate capacitor having square plates 4.50 cm on each side and 8.00 mm apart is placed in series with the following: an ac source of angular frequency 650 rad/s and voltage amplitude 22.5 V; a 75.0 Ω resistor; and an ideal solenoid that is 9.00 cm long, has a circular cross section 0.500 cm in diameter, and carries 125 coils per centimeter. What is the resonance angular frequency of this circuit? (See Exercise 30.11.)

31.42 •• **CP** A toroidal solenoid has 3000 closely wound turns, cross-sectional area 0.500 cm^2, mean radius 9.25 cm, and resistance $R = 2.80\ \Omega$. Ignore the variation of the magnetic field across the cross section of the solenoid. What is the amplitude of the current in the solenoid if it is connected to an ac source that has voltage amplitude 23.5 V and frequency 375 Hz?

31.43 •• A series circuit has an impedance of 58.0 Ω and a power factor of 0.720 at 45.0 Hz. The source voltage lags the current. (a) What circuit element, an inductor or a capacitor, should be placed in series with the circuit to raise its power factor? (b) What size element will raise the power factor to unity?

31.44 •• Consider the ac adapter that bisects the power cord to your laptop computer, cell phone charger, or another electronic device. (a) What dc output voltage V and current I are listed on your device? (b) What power P is listed on your device? Does $P = IV$? (c) Your ac adapter is most likely not a simple transformer; however, it is instructive to analyze the device as if it were: Assume your device is a transformer with a 230 V rms voltage on its primary coil and a full-wave rectifier following its secondary coil. If an *R-C* circuit with a sufficient time constant followed, the ultimate output would maintain the dc value of the voltage amplitude on the rectified secondary coil. If the primary coil has 200 turns, how many turns are needed on the secondary coil to produce the stated dc output according to this scheme? (Give the closest integer value.) (d) Estimate the current amplitude in the primary coil. (e) Based on the size of your device and assuming the transformer geometry shown in **Fig. P31.44**, estimate the length l of the primary coil. (f) Use Ampere's law to estimate the average strength of the magnetic field inside the core, assuming a relative permeability of 5000.

Figure **P31.44**

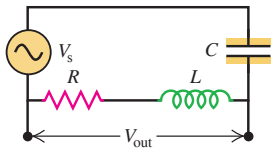


31.45 •• In an L-R-C series circuit, $R = 380\ \Omega$, $X_C = 360\ \Omega$, and $X_L = 560\ \Omega$. The average electrical power consumed in the resistor is 60.0 W. (a) What is the power factor of the circuit? (b) What is the rms voltage of the source?

31.46 • At a frequency ω_1 the reactance of a certain capacitor equals that of a certain inductor. (a) If the frequency is changed to $\omega_2 = 5\omega_1$, what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (b) If the frequency is changed to $\omega_3 = \omega_1/7$, what is the ratio of the reactance of the inductor to that of the capacitor? Which reactance is larger? (c) If the capacitor and inductor are placed in series with a resistor of resistance R to form an L-R-C series circuit, what will be the resonance angular frequency of the circuit?

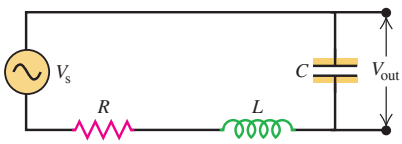
31.47 •• **A High-Pass Filter.** One application of L-R-C series circuits is to high-pass or low-pass filters, which filter out either the low- or high-frequency components of a signal. A high-pass filter is shown in **Fig. P31.47**, where the output voltage is taken across the L-R combination. (The L-R combination represents an inductive coil that also has resistance due to the large length of wire in the coil.) Derive an expression for V_{out}/V_s, the ratio of the output and source voltage amplitudes, as a function of the angular frequency ω of the source. Show that when ω is small, this ratio is proportional to ω and thus is small, and show that the ratio approaches unity in the limit of large frequency.

Figure P31.47



31.48 •• **A Low-Pass Filter. Figure P31.48** shows a low-pass filter (see Problem 31.47); the output voltage is taken across the capacitor in an L-R-C series circuit. Derive an expression for V_{out}/V_s, the ratio of the output and source voltage amplitudes, as a function of the angular frequency ω of the source. Show that when ω is large, this ratio is proportional to ω^{-2} and thus is very small, and show that the ratio approaches unity in the limit of small frequency.

Figure P31.48



31.49 ••• An L-R-C series circuit is connected to an ac source of constant voltage amplitude V and variable angular frequency ω. (a) Show that the current amplitude, as a function of ω, is

$$I = \frac{V}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

(b) Show that the average power dissipated in the resistor is

$$P = \frac{V^2R/2}{R^2 + (\omega L - 1/\omega C)^2}$$

(c) Show that I and P are *both* maximum when $\omega = 1/\sqrt{LC}$, the resonance frequency of the circuit. (d) Graph P as a function of ω for $V = 100$ V, $R = 200\ \Omega$, $L = 2.0$ H, and $C = 0.50\ \mu$F. Compare to the light purple curve in Fig. 31.18. Discuss the behavior of I and P in the limits $\omega = 0$ and $\omega \to \infty$.

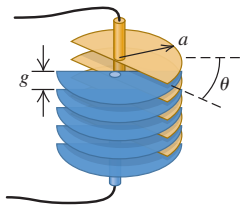
31.50 ••• An L-R-C series circuit is connected to an ac source of constant voltage amplitude V and variable angular frequency ω. Using the results of Problem 31.49, find an expression for (a) the amplitude V_L of the voltage across the inductor as a function of ω and (b) the amplitude V_C of the voltage across the capacitor as a function of ω. (c) Graph V_L and V_C as functions of ω for $V = 100$ V, $R = 200\ \Omega$, $L = 2.0$ H, and $C = 0.50\ \mu$F. (d) Discuss the behavior of V_L and V_C in the limits $\omega = 0$ and $\omega \to \infty$. For what value of ω is $V_L = V_C$? What is the significance of this value of ω?

31.51 •• In an L-R-C series circuit the magnitude of the phase angle is 54.0°, with the source voltage lagging the current. The reactance of the capacitor is 350 Ω, and the resistor resistance is 180 Ω. The average power delivered by the source is 140 W. Find (a) the reactance of the inductor; (b) the rms current; (c) the rms voltage of the source.

31.52 •• CP Cell phones that use 4G technology receive signals broadcast between 2 GHz and 8 GHz. (a) If you want to create a simple L-R-C series circuit to detect a 4.0 GHz cell phone signal, what is the relevant value of the product LC, where L is the inductance and C is the capacitance? (b) If you choose a capacitor that has $C = 1.0 \times 10^{-15}$ F, what inductance do you need? (c) Suppose you want to wind your own toroidal inductor and fit it inside a box as thin as your cell phone. Based on the size of your phone, estimate the largest cross-sectional area possible for this. (d) Assume the largest allowable radius of the toroid is 1.0 cm and estimate the lowest number of windings needed to create your inductor, assuming the material inside has a relative permeability of 1.

31.53 ••• CP Five semicircular conducting plates, each with radius a, are attached to a conducting rod along their common axis such that the plates are parallel with a separation gap g between adjacent plates. A similar arrangement involving five additional semicircular conducting plates is attached to a second conducting rod symmetrically intertwined, upside down, with the first arrangement such that the plates can rotate freely, as shown in **Fig. P31.53**. The two five-plate constructions, with an adjacent plate separation of $g/2$, are insulated from each other, and wire leads are attached to the separate axles. This is a common type of variable capacitor, with a capacitance proportional to the overlap angle θ. (a) Determine a formula for the capacitance C in terms of a, g, and θ, where θ is in radians. (b) The plate radius is $a = 6.00$ cm while the gap distance is $g = 2.00$ mm. This variable capacitor is placed in series with a 100 μH inductor and a 10.0 Ω resistor. A voltage source is supplied by an amplified signal from an antenna, and the output voltage is taken across the capacitor. The circuit is used as a radio receiver that receives a signal frequency equal to the resonance frequency of the L-R-C circuit. To receive an AM radio signal at 1180 kHz, what angle θ, in degrees, should be selected? (c) If the input signal is a sinusoidal voltage with amplitude 100.0 mV and frequency 1180 kHz, what is the amplitude of the output voltage? (d) If the knob on the capacitor is turned so that $\theta = 120°$, to what frequency is the circuit tuned? (e) If the input signal has the frequency resonant with this setting and voltage amplitude 100 mV, what is the voltage amplitude of the output signal?

Figure P31.53



31.54 •• **The L-R-C Parallel Circuit.** A resistor, an inductor, and a capacitor are connected in parallel to an ac source with voltage amplitude V and angular frequency ω. Let the source voltage be given by $v = V\cos\omega t$. (a) Show that each of the instantaneous voltages v_R, v_L, and v_C at any instant is equal to v and that $i = i_R + i_L + i_C$, where i is the current through the source and i_R, i_L, and i_C are the currents through the resistor, inductor, and capacitor, respectively. (b) What are the phases of i_R, i_L, and i_C with respect to v? Use current phasors to represent i, i_R, i_L, and i_C. In a phasor diagram, show the phases of these four currents with respect to v. (c) Use the phasor diagram of part (b) to show that the current amplitude I for the current i through the source is $I = \sqrt{I_R^2 + (I_C - I_L)^2}$. (d) Show that the result of part (c) can be written as $I = V/Z$, with $1/Z = \sqrt{(1/R^2) + [\omega C - (1/\omega L)]^2}$.

31.55 •• The impedance of an *L-R-C* parallel circuit was derived in Problem 31.54. (a) Show that at the resonance angular frequency $\omega_0 = 1/\sqrt{LC}$, the impedance Z is a maximum and therefore the current through the ac source is a minimum. (b) A 100 Ω resistor, a 0.100 μF capacitor, and a 0.300 H inductor are connected in parallel to a voltage source with amplitude 230 V. What is the resonance angular frequency? For this circuit, what is (c) the maximum current through the source at the resonance frequency; (d) the maximum current in the resistor at resonance; (e) the maximum current in the inductor at resonance; (f) the maximum current in the branch containing the capacitor at resonance?

31.56 •• A 400 Ω resistor and a 6.00 μF capacitor are connected in parallel to an ac generator that supplies an rms voltage of 180 V at an angular frequency of 360 rad/s. Use the results of Problem 31.54. Note that since there is no inductor in this circuit, the $1/\omega L$ term is not present in the expression for $1/Z$. Find (a) the current amplitude in the resistor; (b) the current amplitude in the capacitor; (c) the phase angle of the source current with respect to the source voltage; (d) the amplitude of the current through the generator. (e) Does the source current lag or lead the source voltage?

31.57 ••• An *L-R-C* series circuit consists of a 2.60 μF capacitor, a 6.00 mH inductor, and a 90.0 Ω resistor connected across an ac source of voltage amplitude 15.0 V having variable frequency. (a) Under what circumstances is the average power delivered to the circuit equal to $\frac{1}{2}V_{rms}I_{rms}$? (b) Under the conditions of part (a), what is the average power delivered to each circuit element and what is the maximum current through the capacitor?

31.58 •• An *L-R-C* series circuit has $R = 60.0\ \Omega$, $L = 0.800$ H, and $C = 3.00 \times 10^{-4}$ F. The ac source has voltage amplitude 90.0 V and angular frequency 120 rad/s. (a) What is the maximum energy stored in the inductor? (b) When the energy stored in the inductor is a maximum, how much energy is stored in the capacitor? (c) What is the maximum energy stored in the capacitor?

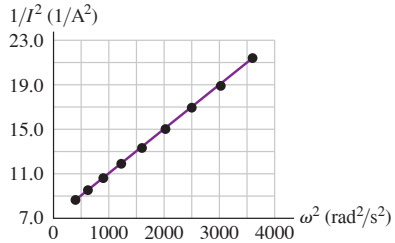
31.59 • In an *L-R-C* series circuit, the source has a voltage amplitude of 230 V, $R = 140.0\ \Omega$, and the reactance of the capacitor is 625 Ω. The voltage amplitude across the capacitor is 575 V. (a) What is the current amplitude in the circuit? (b) What is the impedance? (c) What two values can the reactance of the inductor have? (d) For which of the two values found in part (c) is the angular frequency less than the resonance angular frequency? Explain.

31.60 •• In an *L-R-C* series ac circuit, the source has a voltage amplitude of 230 V, $R = 100.0\ \Omega$, and the reactance of the inductor is 320 Ω. The voltage amplitude across the resistor is 115 V. (a) What is the current amplitude in the circuit? (b) What is the voltage amplitude across the inductor? (c) What two values can the reactance of the capacitor have? (d) For which of the two values found in part (c) is the angular frequency less than the resonance angular frequency? Explain.

31.61 • A resistance R, capacitance C, and inductance L are connected in series to a voltage source with amplitude V and variable angular frequency ω. If $\omega = \omega_0$, the resonance angular frequency, find (a) the maximum current in the resistor; (b) the maximum voltage across the capacitor; (c) the maximum voltage across the inductor; (d) the maximum energy stored in the capacitor; (e) the maximum energy stored in the inductor. Give your answers in terms of R, C, L, and V.

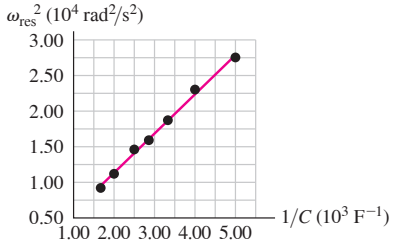
31.62 •• **DATA** A coworker of yours was making measurements of a large solenoid that is connected to an ac voltage source. Unfortunately, she left for vacation before she completed the analysis, and your boss has asked you to finish it. You are given a graph of $1/I^2$ versus ω^2 (**Fig. P31.62**), where I is the current in the circuit and ω is the angular frequency of the source. A note attached to the graph says that the voltage amplitude of the source was kept constant at 12.0 V. Calculate the resistance and inductance of the solenoid.

Figure **P31.62**



31.63 •• **DATA** You are analyzing an ac circuit that contains a solenoid and a capacitor in series with an ac source that has voltage amplitude 90.0 V and angular frequency ω. For different capacitors in the circuit, each with known capacitance, you measure the value of the frequency ω_{res} for which the current in the circuit is a maximum. You plot your measured values on a graph of ω_{res}^2 versus $1/C$ (**Fig. P31.63**). The maximum current for each value of C is the same, you note, and equal to 4.50 A. Calculate the resistance and inductance of the solenoid.

Figure **P31.63**



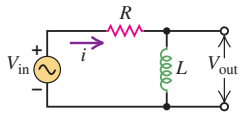
31.64 •• **DATA** You are given this table of data recorded for a circuit that has a resistor, an inductor with negligible resistance, and a capacitor, all in series with an ac voltage source:

f (Hz)	80	160
Z (Ω)	15	13
ϕ (deg)	−71	67

Here f is the frequency of the voltage source, Z is the impedance of the circuit, and ϕ is the phase angle. (a) Use the data at both frequencies to calculate the resistance of the resistor. Calculate the average of these two values of the resistance, and use the result as the value of R in the rest of the analysis. (b) Use the data at 80 Hz and 160 Hz to calculate the inductance L and capacitance C of the circuit. (c) What is the resonance frequency for the circuit, and what are the impedance and phase angle at the resonance frequency?

31.65 ••• **CP** The frequency response of a filter circuit is commonly specified, in decibels, as the attenuation $G = (20\text{ dB})\log_{10}(V_{out}/V_{in})$, where V_{in} is the input voltage amplitude and V_{out} is the output voltage amplitude. This is analogous to the definition of decibels given in Eq. (16.15). For the *R-L* circuit shown in **Fig. P31.65**, the input signal is $v_{in}(t) = V_{in}\cos(\omega t)$. (a) The current is $i(t) = I\cos(\omega t - \phi)$. What is the current amplitude I? (b) What is the phase angle ϕ? (c) What is the ratio V_{out}/V_{in}? (d) For what frequency $f = f_{3db}$ does this circuit have an attenuation of −3.0 dB? (The frequency f_{3db} is commonly cited as the nominal boundary demarcating the "pass band.") (e) If $R = 100\ \Omega$, what inductance L will "pass" frequencies higher than 10.0 kHz and block frequencies lower than 10.0 kHz?

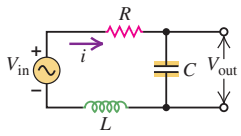
Figure **P31.65**



CHALLENGE PROBLEMS

31.66 ••• CALC The *L-R-C* series circuit shown in **Fig. P31.66** has an ac voltage source $v_{in}(t) = V_{in}\cos(\omega t)$, where V_{in} is the input voltage amplitude and ω is its angular frequency. The values of the resistance, inductance, and capacitance of the components are *R, L*, and *C*. The capacitance is variable. The output voltage is $v_{out}(t) = V_{out}\cos(\omega t + \theta)$. (a) What is the output voltage amplitude V_{out}? (b) What is the phase angle θ of the output voltage? (c) What is the value of θ at resonance? (d) If $R = 100\ \Omega$, $L = 1.00\ \mu$H, and $V_{in} = 10.0$ V, what value of *C* would result in a resonance frequency of 100 kHz? (e) In that case what would be the output voltage amplitude at resonance?

Figure P31.66



31.67 •• CALC In an *L-R-C* series circuit the current is given by $i = I\cos\omega t$. The voltage amplitudes for the resistor, inductor, and capacitor are V_R, V_L, and V_C. (a) Show that the instantaneous power into the resistor is $p_R = V_R I\cos^2\omega t = \frac{1}{2}V_R I(1 + \cos 2\omega t)$. What does this expression give for the average power into the resistor? (b) Show that the instantaneous power into the inductor is $p_L = -V_L I\sin\omega t\cos\omega t = -\frac{1}{2}V_L I\sin 2\omega t$. What does this expression give for the average power into the inductor? (c) Show that the instantaneous power into the capacitor is $p_C = V_C I\sin\omega t\cos\omega t = \frac{1}{2}V_C I\sin 2\omega t$. What does this expression give for the average power into the capacitor? (d) The instantaneous power delivered by the source is shown in Section 31.4 to be $p = VI\cos\omega t(\cos\phi\cos\omega t - \sin\phi\sin\omega t)$. Show that $p_R + p_L + p_C$ equals p at each instant of time.

31.68 ••• CALC (a) At what angular frequency is the voltage amplitude across the *resistor* in an *L-R-C* series circuit at maximum value? (b) At what angular frequency is the voltage amplitude across the *inductor* at maximum value? (c) At what angular frequency is the voltage amplitude across the *capacitor* at maximum value? (You may want to refer to the results of Problem 31.49.)

MCAT-STYLE PASSAGE PROBLEMS

BIO Converting dc to ac. An individual cell such as an egg cell (an ovum, produced in the ovaries) is commonly organized spatially, as manifested in part by asymmetries in the cell membrane. These asymmetries include nonuniform distributions of ion transport mechanisms, which result in a net electric current entering one region of the membrane and leaving another. These steady cellular currents may regulate cell polarity, leading (in the case of eggs) to embryonic polarity; therefore scientists are interested in measuring them.

These cellular currents move in loops through extracellular fluid. Ohm's law requires that there be voltage differences between any two points in this current-carrying fluid surrounding cells. Although the currents may be significant, the extracellular voltage differences are tiny—on the order of nanovolts. If we can map the voltage differences in the fluid outside a cell, we can calculate the current density by using Ohm's law, assuming that the resistivity of the fluid is known. We cannot measure these voltage differences by spacing two electrodes 10 or 20 μm apart, because the dc impedance (the resistance) of such electrodes is high and the inherent noise in signals detected at the electrodes far exceeds the cellular voltages.

One successful method of measurement uses an electrode with a ball-shaped end made of platinum that is moved sinusoidally between two points in the fluid outside a cell. The electric potential that the electrode measures, with respect to a distant reference electrode, also varies sinusoidally. The dc potential difference between the two extremes (the two points in the fluid) is then converted to a sine-wave ac potential difference. The platinum electrode behaves as a capacitor in series with the resistance of the extracellular fluid. This resistance, called the *access resistance* (R_A), has a value of about $\rho/10a$, where ρ is the resistivity of the fluid (usually expressed in $\Omega\cdot$cm) and a is the radius of the ball electrode. The platinum ball typically has a diameter of 20 μm and a capacitance of 10 nF; the resistivity of many biological fluids is 100 $\Omega\cdot$cm.

31.69 What is the dc impedance of the electrode, assuming that it behaves as an ideal capacitor? (a) 0; (b) infinite; (c) $\sqrt{2}\times 10^4\ \Omega$; (d) $\sqrt{2}\times 10^6\ \Omega$.

31.70 If the electrode oscillates between two points 20 μm apart at a frequency of $(5000/\pi)$ Hz, what is the electrode's impedance? (a) 0; (b) infinite; (c) $\sqrt{2}\times 10^4\ \Omega$; (d) $\sqrt{2}\times 10^6\ \Omega$.

31.71 The signal from the oscillating electrode is fed into an amplifier, which reports the measured voltage as an rms value, 1.5 nV. What is the potential difference between the two extremes? (a) 1.5 nV; (b) 3.0 nV; (c) 2.1 nV; (d) 4.2 nV.

31.72 If the frequency at which the electrode is oscillated is increased to a very large value, the electrode's impedance (a) approaches infinity; (b) approaches zero; (c) approaches a constant but nonzero value; (d) does not change.

ANSWERS

Chapter Opening Question ?

(iv) A radio simultaneously detects transmissions at *all* frequencies. However, a radio is an *L-R-C* series circuit, and at any given time it is tuned to have a resonance at just one frequency. Hence the response of the radio to that frequency is much greater than its response to any other frequency, which is why you hear only one broadcasting station through the radio's speaker. (You can sometimes hear a second station if its frequency is sufficiently close to the tuned frequency.)

Key Example VARIATION Problems

VP31.3.1 (a) $1.70\times 10^6\ \Omega$ (b) 2.48×10^{-3} A
VP31.3.2 (a) $(0.300\text{ V})\cos[(1.75\times 10^3\text{ rad/s})t]$ (b) 81.6 Ω (c) $(0.196\text{ V})\cos[(1.75\times 10^3\text{ rad/s})t - \pi/2\text{ rad}]$
VP31.3.3 (a) 128 Hz (b) both are 0.620 V (c) zero; zero
VP31.3.4 (a) 0.0215 A (b) 0.0116 A (c) 2.45 V (d) 1.72 μF
VP31.5.1 (a) 902 Hz (b) 316 Ω (c) 0.174 A (d) –36.1° (e) lags
VP31.5.2 (a) 387 Ω (b) 0.0422 A (c) 16.3 V (d) 1.48 V (e) 13.0 V
VP31.5.3 (a) 17.4 Ω (b) 8.65 mH
VP31.5.4 (a) $V_R = 11.4$ V, $V_L = 6.24$ V, $V_C = 34.1$ V (b) $v_R = -8.71$ V, $v_L = -4.03$ V, $v_C = +22.0$ V
VP31.7.1 (a) 950 W (b) 1900 W (c) 55.7 Ω
VP31.7.2 (a) $X_L = 107\ \Omega$, $X_C = 699\ \Omega$ (b) –65.1° (c) 0.421
VP31.7.3 (a) 653 Ω (b) 0.0536 A (c) 0.395 W
VP31.7.4 (a) $X_L = 97.5\ \Omega$, $X_C = 85.5\ \Omega$ (b) 36.9° (c) 16.0 Ω

Bridging Problem

(a) 8.35×10^4 rad/s and 3.19×10^5 rad/s
(b) At 8.35×10^4 rad/s: $V_{source} = 49.5$ V, $I = 0.132$ A, $V_R = 16.5$ V, $V_L = 16.5$ V, $V_C = 63.2$ V
At 3.19×10^5 rad/s: $V_{source} = 49.5$ V, $I = 0.132$ A, $V_R = 16.5$ V, $V_L = 63.2$ V, $V_C = 16.5$ V